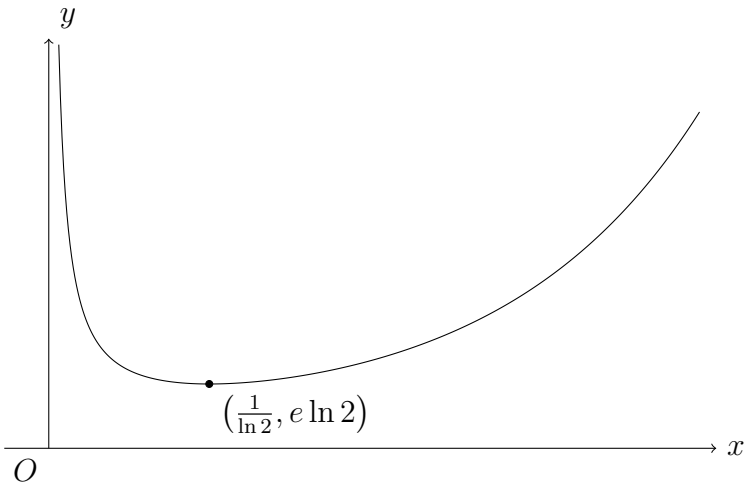
(i) Let $u = t - 1$

$$f(x) = \int_0^2 u^{x-1}\,du = \left[\frac{u^x}{x}\right]_0^2 = \frac{2^x}{x}$$

$$\frac{dy}{dx} = \frac{x2^x \ln 2 - 2^x}{x^2}$$

$$\frac{dy}{dx} = 0 \implies x = \frac{1}{\ln 2}$$

x	0^+	$\frac{1}{\ln 2}$	∞
y	∞	$e \ln 2$	∞



(ii)

$$g(x) = \int_{-1}^{1} \frac{1}{\sqrt{1 - 2xt + x^2}}\,dt$$

For $x \neq 0$, For $x \neq 0$,
Let $u = 1 - 2xt + x^2$, $\dfrac{du}{dt} = -2x$

$$\begin{aligned}\int \frac{1}{\sqrt{1 - 2xt + x^2}}\,dt &= \int u^{-\frac{1}{2}} \cdot \left(-\frac{1}{2x}\right)\,du \\ &= -\frac{1}{x}u^{\frac{1}{2}} \\ &= -\frac{1}{x}\sqrt{1 - 2xt + x^2}\end{aligned}$$

$$\therefore g(x) = \left[-\frac{1}{x}\sqrt{1-2xt+x^2}\right]_{-1}^{1}$$

For $x = 0$,

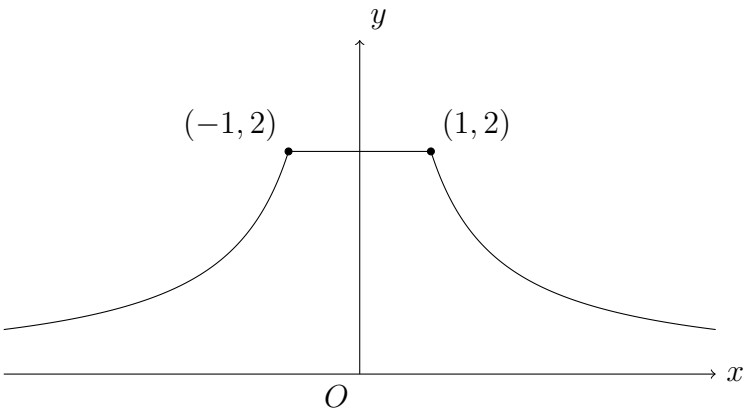
$$g(0) = \int_{-1}^{1} 1\,dt = 2$$

$$\begin{aligned} g(x) &= \left[-\frac{1}{x}\sqrt{1-2xt+x^2}\right]_{-1}^{1} \\ &= \frac{1}{x}\left(\sqrt{x^2+2x+1} - \sqrt{x^2-2x+1}\right) \\ &= \frac{1}{x}\left(|x+1| - |x-1|\right) \end{aligned}$$

$$g(0) = \int_{-1}^{1} 1\,dt = 2$$

$$g(x) = \begin{cases} -\dfrac{2}{x}, & x < -1, \\ 2, & -1 \le x \le 1, \\ \dfrac{2}{x}, & x > 1. \end{cases}$$

x	$-\infty$	-1	0	1	∞
y	0^+	2	2	2	0^+



This solution was submitted by **Jon**

Published and refined by **Jon**

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